Farmers’ Smile One Page Technical Report

Prototype V1.0 | Agriculture | Small AI for Development

1 Challenge Tackled

Farmers in Thailand must consider what to plant when rainfall and water access are uncertain. Farmers’ Smile is a mobile web guide for Chiang Mai, Songkhla and Surat Thani. A farmer chooses a province, planting month and water option, then sees up to three crop ideas with plain-language reasons. The design supports Thai and English use and limited connectivity.

2 Tools / ML Models Used

- **Data.** NASA POWER daily temperature and rainfall, 1996–2025, at one reference grid per province; Python scripts prepare weather summaries and local model inputs.
- **Model and rules.** A local seven-neighbor historical-season matcher (k = 7) retrieves similar past seasons; separate crop temperature and water rules rank options and generate cautions.
- **App.** HTML, CSS, JavaScript and bundled Leaflet with geoBoundaries geometry; a downloadable standalone HTML includes data, map and crop photos. AI tools helped us write code; using the app makes no paid language-model call.

3 What Worked Well

- **Checks.** All 72 province × month × water combinations passed the available engine checks. The interface presents only the top three climate-fit options, one-to-five stars and short explanations.
- **Data and offline.** Each area has 30 complete years of prepared daily history (10,958 records). In a local Chromium check, the saved offline HTML opened and returned crop results without network requests.

4 What Was Challenging

- **Map.** Province outlines and a clean Thailand-only map required grouping administrative geometry correctly; only the three pilot provinces are selectable.
- **Offline and limits.** Bundling map geometry, data, code and crop photos yielded one file of about 4.19 MB. Coarse weather grids and broad crop rules remain limits: no soil, farm yields or farmer trials validate the rankings. Stars express relative historical climate fit, not a next-year forecast or success probability.

5 How You Spent Your Time

Work sequence: scope the farmer decision and pilot provinces; source and prepare data; build the local comparison and mobile map; package the offline copy; test inputs and prepare communication. Shine led research, product design and AI-assisted prototyping; Chan and Muay documented and presented the work; James edited the video. The prototype was developed iteratively before the event; exact 24-hour time allocations were not recorded.

Method at a glance: province, month and water → seven similar past seasons → fixed crop rules → top three climate-fit options and reasons.

Next: Test wording, ratings and onboarding with farmers and agricultural advisers; compare guidance with field conditions before using it for planting decisions.

Sources: NASA POWER daily API; FAO crop water guide and Ecocrop; geoBoundaries Thailand. Full URLs, credits, methods and run steps are in the project README and SOURCES files.